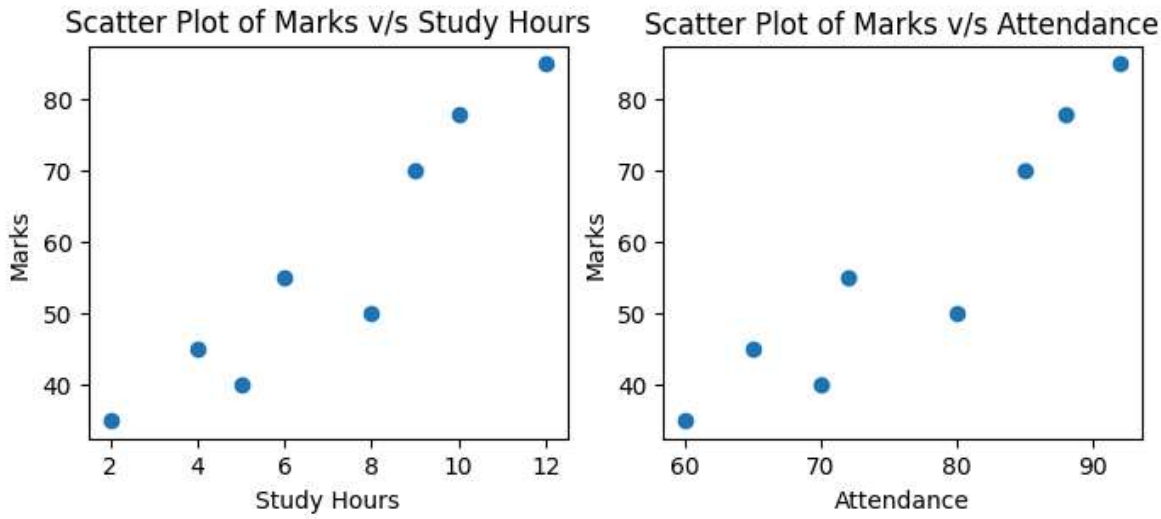
Anlea Maria Jose, 02, Experiment 8

Date: 10-04-2026

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Study Hours': [2,4,5,6,8,9,10,12],
    'Marks': [35,45,40,55,50,70,78,85],
    'Attendance': [60,65,70,72,80,85,88,92]
}

df = pd.DataFrame(data)
print("Dataset:\n")
print(df)

plt.figure(figsize=(8,3))
plt.subplot(121)
plt.scatter(df['Study Hours'], df['Marks'])
plt.xlabel('Study Hours')
plt.ylabel('Marks')
plt.title('Scatter Plot of Marks v/s Study Hours')

plt.subplot(122)
plt.scatter(df['Attendance'], df['Marks'])
plt.xlabel('Attendance')
plt.ylabel('Marks')
plt.title('Scatter Plot of Marks v/s Attendance')
plt.show()
```

```
Dataset:

   Study Hours  Marks  Attendance
0            2     35          60
1            4     45          65
2            5     40          70
3            6     55          72
4            8     50          80
5            9     70          85
6           10     78          88
7           12     85          92
```



```
# Spearman's Rank Correlation

df['Rank_X'] = df['Study Hours'].rank()
df['Rank_Y'] = df['Marks'].rank()

df['d'] = df['Rank_X'] - df['Rank_Y']
df['d^2'] = df['d']**2
print(df)

n = len(df)
sum_d2 = df['d^2'].sum()

spearman_corr = 1 - (6 * sum_d2) / (n * (n**2 - 1))

print("\nSpearman's Rank Correlation: ")
print("Sum of d^2 = ", sum_d2)
print("R = ", round(spearman_corr, 4))

from scipy.stats import spearmanr
spearman_corr_lib = spearmanr(df['Study Hours'], df['Marks'])
print("\nSpearman's Rank Correlation Using Library: ")
print("R_lib = ", round(spearman_corr_lib.correlation, 4))
```

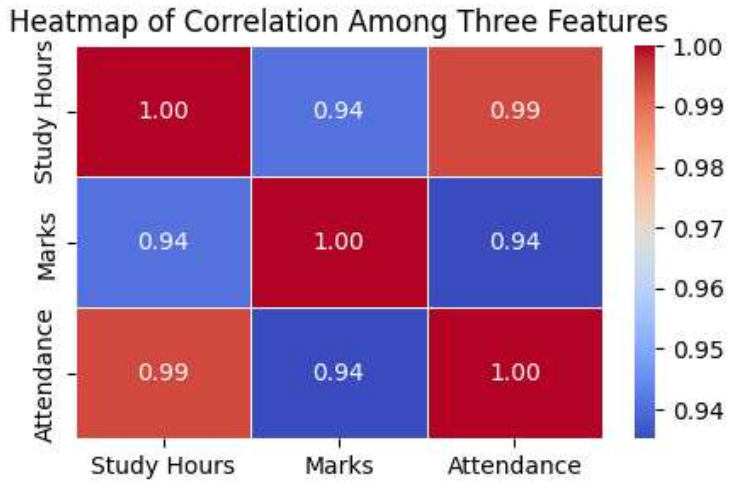
```
   Study Hours  Marks  Attendance  Rank_X  Rank_Y    d  d^2
0            2     35          60     1.0     1.0  0.0  0.0
1            4     45          65     2.0     3.0 -1.0  1.0
2            5     40          70     3.0     2.0  1.0  1.0
3            6     55          72     4.0     5.0 -1.0  1.0
4            8     50          80     5.0     4.0  1.0  1.0
5            9     70          85     6.0     6.0  0.0  0.0
6           10     78          88     7.0     7.0  0.0  0.0
7           12     85          92     8.0     8.0  0.0  0.0

Spearman's Rank Correlation:
Sum of d^2 =  4.0
R =  0.9524

Spearman's Rank Correlation Using Library:
R_lib =  0.9524
```

```
import math
from scipy.stats import pearsonr

df = pd.DataFrame(data)
x = df['Study Hours']
y = df['Marks']

x_bar = x.mean()
y_bar = y.mean()

df['x - x_bar'] = x - x_bar
df['y - y_bar'] = y - y_bar

df['(x - x_bar)(y - y_bar)'] = df['x - x_bar'] * df['y - y_bar']
df['(x - x_bar)^2'] = (df['x - x_bar'])**2
df['(y - y_bar)^2'] = (df['y - y_bar'])**2
print(df)

sum_xy = df['(x - x_bar)(y - y_bar)'].sum()
sum_x2 = df['(x - x_bar)^2'].sum()
sum_y2 = df['(y - y_bar)^2'].sum()

pearson_corr = sum_xy / math.sqrt(sum_x2 * sum_y2)

print("\nKarl Pearson's Coefficient of Correlation Manually")
```

```
print("Sum of X = ", round(x_bar, 4))
print("Sum of Y = ", round(y_bar, 4))
print("Sum of (x-x_bar)(y-y_bar)", round(sum_xy, 4))
print("Sum of (x-x_bar)^2", round(sum_x2, 4))
print("Sum of (y-y_bar)^2", round(sum_y2, 4))
print("R = ", round(pearson_corr, 4))
```

```
   Study Hours  Marks  Attendance  x - x_bar  y - y_bar  \
0            2     35          60       -5.0     -22.25
1            4     45          65       -3.0     -12.25
2            5     40          70       -2.0     -17.25
3            6     55          72       -1.0      -2.25
4            8     50          80        1.0      -7.25
5            9     70          85        2.0      12.75
6           10     78          88        3.0      20.75
7           12     85          92        5.0      27.75

   (x - x_bar)(y - y_bar)  (x - x_bar)^2  (y - y_bar)^2
0                  111.25           25.0       495.0625
1                   36.75            9.0       150.0625
2                   34.50            4.0       297.5625
3                    2.25            1.0         5.0625
4                   -7.25            1.0        52.5625
5                   25.50            4.0       162.5625
6                   62.25            9.0       430.5625
7                  138.75           25.0       770.0625

Karl Pearson's Coefficient of Correlation Manually
Sum of X =  7.0
Sum of Y =  57.25
Sum of (x-x_bar)(y-y_bar) 404.0
Sum of (x-x_bar)^2 78.0
Sum of (y-y_bar)^2 2363.5
R =  0.9409
```

```
#Heatmap for Three Features
import matplotlib.pyplot as plt
import seaborn as sns
import math
from scipy.stats import spearmanr, pearsonr

# Function to calculate Pearson correlation
def pearson(a, b):
    mean_a = sum(a) / len(a)
    mean_b = sum(b) / len(b)

    num = 0
    den_a = 0
    den_b = 0

    for i in range(len(a)):
        da = a[i] - mean_a
        db = b[i] - mean_b
        num += da * db
        den_a += da ** 2
        den_b += db ** 2


    return num / math.sqrt(den_a * den_b)

# Calculate pairwise correlations manually
df = pd.DataFrame(data)

r11 = pearson(df['Study Hours'], df['Study Hours'])
r22 = pearson(df['Marks'], df['Marks'])
r33 = pearson(df['Attendance'], df['Attendance'])
```

```
r12 = pearson(df['Study Hours'], df['Marks'])
r13 = pearson(df['Study Hours'], df['Attendance'])
r23 = pearson(df['Marks'], df['Attendance'])


# Create correlation matrix manually
corr_matrix = pd.DataFrame([
    [r11,   r12, r13],
    [r12, r22,   r23],
    [r13, r23, r33]
], columns=['Study Hours', 'Marks', 'Attendance'],
   index=['Study Hours', 'Marks', 'Attendance'])

print("\nCorrelation Matrix for Heatmap")
print(corr_matrix)


#Correlation matrix using library
print("\nCorrelation Matrix using library")
corr_matrix_lib = df.corr()
print(corr_matrix_lib)


# Plot heatmap
plt.figure(figsize=(5, 3))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Heatmap of Correlation Among Three Features")
plt.show()
```

```
Correlation Matrix for Heatmap
             Study Hours     Marks  Attendance
Study Hours     1.000000  0.940928    0.994553
Marks           0.940928  1.000000    0.935178
Attendance      0.994553  0.935178    1.000000

Correlation Matrix using library
             Study Hours     Marks  Attendance
Study Hours     1.000000  0.940928    0.994553
Marks           0.940928  1.000000    0.935178
Attendance      0.994553  0.935178    1.000000
```